Shallow Neural Network-Based Error Compensation of a Non-Linear SAR ADC Sampling Stage

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***Abstract*—Non-idealities of the track-and-hold (T&H) circuit in analog-to-digital converters (ADCs) can severely limit the ADC linearity. In practical applications, traditional calibration techniques are ineffective in suppressing these effects due to the multi-dimensional dependency of the non-idealities on e.g., input voltage, frequency and temperature. To address this, we propose a calibration solution that combines circuit knowledge of the error mechanisms with a data-driven approach into a shallow neural network (NN) containing just 5 neurons. Due to the small size of the NN, the output of the network can be interpreted and referred back to the underlying circuit non-idealities improving the predictability and reliability of the model. The effectiveness of the NN-based calibration is verified on silicon measurements of 10 different 12-bit SAR ADC samples manufactured in a 65nm CMOS technology, demonstrating an average improvement up to 6.8dB and 13.3dB on the measured SNDR and SFDR, respectively. To the best of the author's knowledge, this is the first demonstration of a compact, interpretable NN-based calibration solution tested over amplitude, frequency, and temperature for multiple samples, while maintaining physics-based insight into the correction process.**

***Index Terms*—Analog-to-digital converter, calibration, neural network, interpretable machine learning, error post-correction.**

I. Introduction

ADCs are an essential building block for a wide range of applications, such as wireless communication and instrumentation. These applications have an increasing need for high sampling rate and high-linearity ADCs. In particular, the T&H switch can have a significant impact on the achievable linearity of the ADC as it can suffer from various analog impairments [1] such as signal-dependent charge injection, leakage and non-linear on-resistance of the sampling switch.

To enhance the linearity of the T&H switch, analog enhancements [2], [3] and digital compensation schemes [4] have been introduced. The latter benefit from the progress made in technology scaling, enabling increasingly more complex processing capabilities in the digital domain. This gave rise to new paradigms such as digitally-assisted analog design, which has been extensively used in the past as a means to lower power dissipation and alleviate design trade-offs in the design of ADCs [5]. While several studies have been conducted on modeling the non-linearity of the T&H switch [6], [7], none of the approaches captures the full complexity of the error mechanism which should include e.g., its dependency on the operating temperature. Besides, the effectiveness of calibration methods targeting a specific error type can be largely reduced when multiple error mechanisms are simultaneously present. To the best of the author's knowledge, no prior art has demonstrated the calibration of these complex T&H switch non-linearities over different operating temperatures and verified it on silicon measurement data.

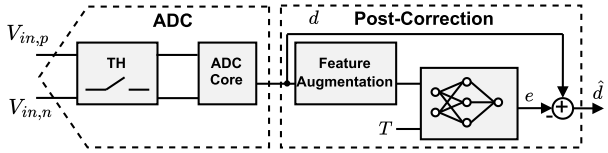


Fig. 1. Simplified schematic of an ADC with a digital neural network-based post correction module, where T is a digital signal representing the on-chip temperature.

In this paper, a hybrid compensation scheme in the form of a domain knowledge-infused neural network is presented that compensates the T&H switch non-linearity at different operating temperatures and is validated on silicon measurement data of multiple ADC samples over a range of input amplitudes and input frequencies. In Fig. 1, the main idea of the proposed technique is shown: the distorted signal d that is produced by the ADC is augmented and fed into a neural network alongside the on-chip temperature T. The neural network then computes a correction value e that compensates for the temperature-dependent distortion which is subtracted from the ADC output to create a linearized output $\hat{d}$.

The main contributions in this paper are threefold: first, it is shown that multiple non-linearity mechanisms of the T&H switch can simultaneously be corrected including the temperature dependency of those mechanisms; secondly, it is shown that by infusing the neural network with domain knowledge and exploiting structural features (e.g. symmetry), the network can be pruned to just 5 neurons; thirdly, the learned features of the NN trained on silicon data are shown to have strong structural similarity with the simulation data of the T&H circuit, validating the ability of the NN to model

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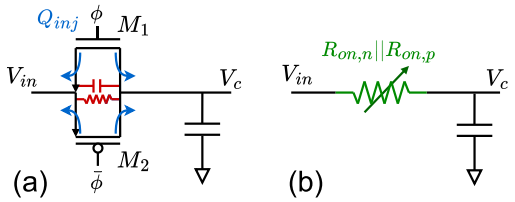


Fig. 2. (a) Single-ended T&H CMOS switch with charge injection (blue) at the track to hold transition, capacitive coupling and leakage (red) during the hold phase. (b) The equivalent on-resistance during the tracking phase.

the physical errors. The calibration method is verified on silicon measurement data from 10 different samples of a 12-bit SAR ADC produced in a 65nm CMOS technology that utilizes a differential CMOS T&H Switch (Fig. 2) running at a sampling rate of 400kS/s and a reduced supply voltage of 1.0V. By reducing the supply voltage, the dynamic errors – which are usually dominating linearity degradation of the T&H circuit – become apparent at lower sampling and input frequencies without altering their underlying mechanisms.

The rest of the paper is organized as follows. Section II discusses the error mechanisms present in T&H switches and Section III describes the proposed neural network architecture. The results are presented in Section IV after which the internal behavior of the NN is analyzed in Section V and an implementation is given in Section VI. The paper is concluded in Section VII and VIII.

II. Sampler Non-Linearity

The T&H switch in an ADC can be the limiting factor for the overall converter linearity, and can be particularly hard to model since the input is a continuous-time signal, while the output is discrete time, and a variety of static and dynamic error sources are present, that are dependent on the operational conditions (e.g., temperature). The non-linearity in the sampler is caused by different error mechanisms [7] such as signal-dependent on-resistance, charge injection, aperture error, capacitive coupling and transistor leakage.

The complexity arises from the fact that these errors are simultaneously present, and the different dependencies of each error mechanism on the input signal amplitude, input signal frequency, and transistor parameters such as the threshold voltage and mobility. Additionally, some of the transistor parameters such as the threshold voltage and leakage current are influenced by the operating temperature of the IC, further complicating the error models. While it is beyond the scope of this paper to analyze and model all individual error sources, for the sake of illustration, two of the most dominant error sources, being the non-linear on-resistance and charge injection, will be discussed to highlight their signal and temperature dependencies.

In Fig. 2, a simplified overview of the sampling stage of the ADC used in this work is shown; each branch of the differential sampling stage contains two complementary MOS transistors (M_1, M_2) that are driven by a complementary clock signal on the gate terminal. During the tracking phase, the equivalent on-resistance of M_1 and M_2 is modulated by the input signal through the non-constant overdrive voltage and the body effect. The equivalent on-resistance of M_1 is given by [8]:

$$R_{on,n} = \frac{L}{\mu_n C_{ox} W(V_{DD} - V_{in} - V_{th})} \tag{1}$$

where

$$V_{th} = V_{th0} + \gamma(\sqrt{2\psi_F + V_{sb}} - \sqrt{2\psi_F}) \tag{2}$$

and

$$\psi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) \tag{3}$$

where V_{th0} denotes the zero substrate bias threshold, $2\psi_F$ the surface potential, γ the body effect parameter, T the temperature in Kelvin, k the Boltzmann constant, q the elementary charge, N_A and n_i the doping and intrinsic doping parameter of the transistor substrate respectively. Whereas the non-linear on-resistance is a memoryless phenomenon, the interaction with the sampling capacitor introduces a dynamic error. The interaction of the non-linear on-resistance and the sampling capacitance can be described by a first-order differential equation:

$$V_c = V_{in} - (R_{on,n}||R_{on,p}) \cdot C\frac{dV_c}{dt} \tag{4}$$

where V_c is the voltage over the sampling capacitor which serves as input signal to the comparator of the SAR ADC and $R_{on,p}$ is the on-resistance of M_2 which is similar to (1) but complementary. The relation between the non-linear on-resistance and the output voltage error can be approximated using a polynomial equation [9]. Assuming that the bandwidth of the sampling switch is sufficiently high, that V_c can closely track V_{in}, and furthermore assuming that the ADC has sufficient resolution and linearity, i.e., V_{out}, which is the analog equivalent of the digital output code of the ADC closely resembles V_c, (4) can be approximated by:

$$\begin{aligned} &V_{in}[n] \\ &= (c_1 + c_2 V_{out}[n] + c_3 V_{out}[n]^2 + c_4 V_{out}[n]^3 + \ldots) \cdot V'_{out}[n]. \end{aligned} \tag{5}$$

Note that although bootstrapping [2] is a proven technique to reduce the non-linear on-resistance of the sampling switch, it does not eliminate this non-linearity, as the body effect still modulates the threshold voltage.

Besides the tracking phase, at the transition to the hold phase, errors due to charge injection are introduced. At the transition, the toggling of the clock signal will cause the charge in the channel to be released; the amount of charge depends on the overdrive voltage in the on-state thus is input signal and temperature dependent [7]. Contrary to the non-linear on-resistance, the charge injection is not frequency dependent and is typically not limiting the linearity at higher input frequencies, where the non-linear on-resistance dominates the linearity degradation [9].

One popular approach to model the non-linear behavior of a dynamic error mechanism is the Volterra series. Although efforts towards the reduction of the number of parameters and computations in the Volterra series have been made [10], [11], the computational complexity is often still a concern for on-chip implementation and the calibration methods can impose additional ADC requirements such as oversampling [11]. In [4], an off-chip post-processing method based on an analytic

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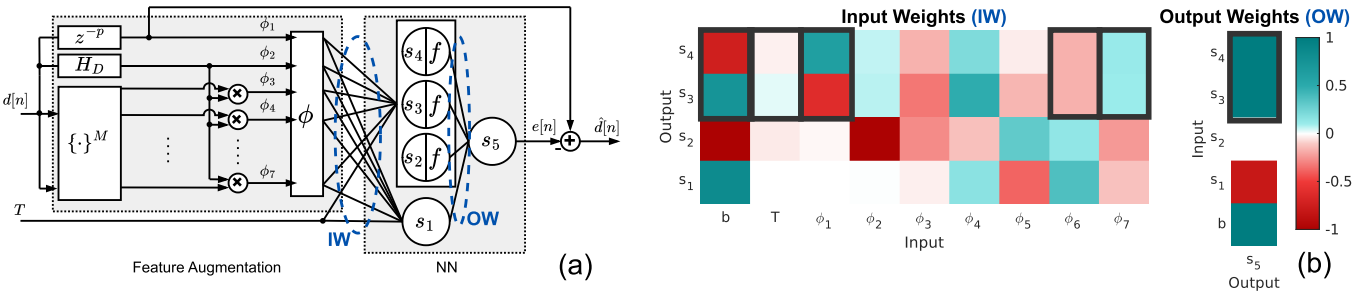


Fig. 3. (a) Block diagram of the feature augmentation and NN architecture, where H_D denotes a digital derivative filter and z^{-p} is a delay line matching the delay of the derivative filter. (b) A symmetry due to the temperature-dependent bias can be seen upon inspecting the weight matrices; the color of the matrix entries denotes the normalized weight or normalized bias.

equation for the non-linear on-resistance induced tracking error is proposed. The coefficients of the analytical equation are recalculated when the input signal frequency changes, restricting deployment in practical operating conditions where input signal frequency content varies. Moreover, the influence of temperature is not taken into account by the calibration scheme. In [9], a calibration method for the non-linear on-resistance of a bootstrapped NMOS sample switch with bottom plate sampling is presented. In this case, the non-linearity source is mostly confined to the body effect. The correction scheme is tested on silicon measurement data of a single sample, however also in this case temperature effects are not taken into account. In [6], a calibration method that compensates the aperture error is described. Due to the finite slew rate of the clock signal, a signal-dependent time shift of the sampling instant will occur. Although the algorithm is verified at different input frequencies, a single input amplitude is used for all tests even though the exact sampling moment is amplitude dependent. In addition, the calibration method is only verified on a single silicon sample and no temperature dependency is included.

Clearly, the approaches presented in literature have focused on single error mechanisms, did not account for the temperature dependency of transistor parameters, e.g., the threshold voltage, and did not verify the presented calibration solutions at different input conditions and for multiple silicon samples. To address this gap in the literature, we propose a data-driven calibration approach that addresses multiple error sources, their temperature dependencies, and is tested on multiple silicon samples to verify its generalizability. The calibration solution can use a single set of coefficients over samples, temperatures, signal amplitudes and signal frequencies without retraining. Moreover, the effectiveness of the technique can be further improved by moving to from batch to per-sample calibration, at the cost of a slight adaptation of the NN parameters.

III. Neural Network Based Post-Correction

Recently, neural networks have gained popularity as a new instrument to compensate ADC non-idealities. Neural networks have been applied to compensate for static error mechanisms [12], dynamic error mechanisms in time-interleaved ADCs [13], [14], [15], pipeline and SAR ADCs [16], [17], [18] and performance optimization for digitally-assisted analog ADCs [19]. Although these methods can significantly improve the linearity of the ADC, the size of these neural networks and their associated hardware cost can be prohibitive for on-chip implementations. Another factor that hinders the adoption of NN-based calibration schemes into products, is their closed box nature which can be undesirable in certain application domains, like safety-critical automotive applications. In [20], Kolmogorov-Arnold Networks (KANs) are proposed as an alternative to multilayer perceptrons, with the goal of enhancing interpretability. While a compact solution was achieved, the learned features were not analyzed, leaving the connection to the underlying physical problem unexplored.

Contrary to prior art, in this work, the neural network is not treated as a closed box. Instead, the learned features of the NN are analyzed leading to two key advantages: first, the learned features can reveal pruning opportunities, resulting in a network with fewer parameters; secondly, a high-level interpretation of the learned features provides insight into the neural network dynamics, leading to a better understanding of the model and strengthening the confidence in the NN's corrective ability. In Section III-A, it is shown how the input features of the neural network are augmented to reduce the complexity of the NN. Then, in Section III-B, the training setup is shown and the pruning strategy is discussed.

A. Feature Augmentation

In Fig. 3(a), the feature augmentation and NN architecture proposed in this work are depicted. As derived in (5), the dominant error mechanism in the ADC depends on a polynomial expansion of the output code of the ADC and the derivative thereof. Since the neural network in essence learns a non-linear mapping between its input data (ADC output) and the target output data (the error of the ADC output data), the network can be simplified by applying non-linear transformations on the input data when a general function form of the error (e.g. polynomial) is expected. The main advantage is that this reduces the NN complexity leading to easier convergence during training, less chance of overfitting due to the reduced number of parameters and better interpretability. Note that even though the feature augmentation focuses on the dominant error mechanisms, there are no assumptions on the ideality of

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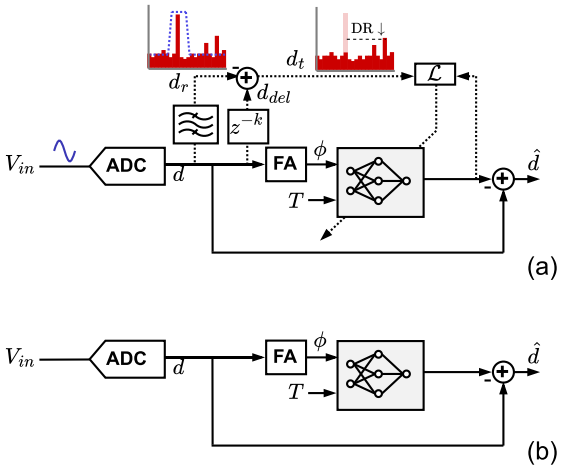


Fig. 4. (a) Training setup, where z^{-k} matches the delay of the bandpass filter to create the target signal. $\mathcal{L}$ denotes the loss function of the neural network. (b) Inference setup.

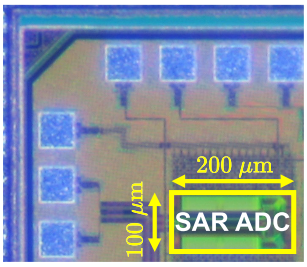


Fig. 5. Die photograph.

the ADC core following the non-linear sampling stage. By performing the experiments on silicon measurement data of multiple dies, the algorithm is shown to function correctly in the presence of other (non-modeled) error sources, thus strengthening the reliability of the proposed method.

Feature augmentation (FA) consists of exponentiation of the digital output code of the ADC, combined with a derivative filter H_D [21] acting on the ADC output which are multiplied to create the input feature vector ϕ together with the on-chip temperature parameter T (Fig. 3(a)). To account for the non-linear features that are dominant in the ADC output data, exponentiation is performed up to the 5th order. For the particular instance depicted in Fig. 3, the features are described by:

$$\phi = [\phi_1, \ldots, \phi_7] = [d, d', d'd, d'd^2, d'd^3, d'd^4, d'd^5] \quad (6)$$

where d is the distorted output signal of the ADC, d' the derivative thereof and the time index n is dropped for the sake of brevity. Note that the input features are split between a signal-dependent part (ϕ) and the temperature (T).

B. Training and Pruning

The augmented input features are fed into the neural network (Fig. 3(a)). The neural network consists of a fully connected hidden layer (s_2, s_3, s_4), a linear neuron that allows input features to directly contribute to the output (s_1), and a linear neuron combining the contributions of the linear and non-linear neurons into a single output value (s_5). The activation function (denoted as f in Fig. 3(a)) used in this network is the hyperbolic tangent, which can be defined as:

$$\tanh(x) = 1 - \frac{2}{e^{2x} + 1}. \quad (7)$$

The hyperbolic tangent is particularly well suited to correct harmonic non-linearities in ADCs since the bias point of the hyperbolic tangent can be controlled to tune the even- and odd-harmonics in a signal, evident from the Taylor expansion of $\tanh(x + b)$ around different bias (b) points.

The neural network including its training process is depicted in Fig. 4(a). First, a known input stimulus V_{in} is applied to the input of the ADC by using a sine wave generator; the ADC then produces the distorted digital output d which is fed through the feature augmentation (FA) block into the NN. To train the NN, a reference signal is needed; the reference signal d_r is obtained by bandpass filtering the distorted signal d, suppressing any other harmonic than the fundamental tone. This removes the need to perform sine-wave fitting on the signal thus simplifying the training setup.

Instead of training the network to directly reconstruct d_r from d, the network is trained on learning the difference between the two signals as this drastically reduces the dynamic range needed by the neural network output, and thus its training time, and increases the correction accuracy. To obtain the target signal d_t, the distorted signal must be delayed to match the bandpass filter delay (z^{-k}). The loss function $\mathcal{L}$ then computes the mean squared error (MSE) between the target signal and the NN output, which is then used to update the weights and biases. After the network is trained using sinusoid input stimuli, the weights are fixed and the network can be used for inference, which is depicted in Fig. 4(b). During inference, the weights are frozen and the NN uses this fixed set of weights to correct all input signals (also signals not present in the training set). Note that the different sinusoid amplitude and frequency combinations used as input to the network during training, combined with the low number of parameters of the network, force the network to generalize, since it does not have sufficient parameters to overfit on all input signals using a fixed set of weights.

After multiple iterations of training and subsequently pruning, the final network architecture and corresponding weights and biases are provided in Fig. 3(a) and Fig. 3(b), respectively. When inspecting the contribution of each neuron to the output value of the NN (s_5), the contribution of neurons s_1, s_3, s_4 were found to be dominant as can be seen from the output weights. During training, the output weights (OWs) connecting neurons s_3 and s_4 to neuron s_5 tended to converge to similar or opposite values; to further analyze this symmetry, these output weights were forced to exactly equal values and the network was retrained. This, in turn, revealed symmetry in the input weights (IWs), where neurons s_3 and s_4 tend to form a pair with similar or opposite weight values (see grouping highlighted in Fig. 3(b) IWs). To confirm these trends, the network was trained multiple times on different samples and

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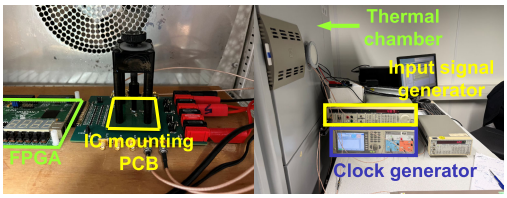


Fig. 6. The measurement setup used for this experiment; the input signal and clock are generated using the Stanford DS360 and Keysight 81134A respectively.

it was verified that it exhibits similar patterns. In Section V, a closer look will be taken at the meaning of these symmetries.

IV. Results

The neural network was verified on silicon measurement data from 12-bit SAR ADC samples produced in 65nm CMOS technology [22]. A die photograph is displayed in Fig. 5. Typically, T&H distortion is an issue for ADCs with high resolution and especially at high frequencies. However, high-frequency signal generators tend to be non-linear and require external filters to improve their performance, which is cumbersome in combination with automated measurements, needed to perform amplitude, frequency, and temperature sweeps. Therefore, the tests in this paper are done at a much lower frequency, allowing the use of a single high precision generator (Stanford DS360) to simplify training and testing. The setup that was used to obtain the measurement data is shown in Fig. 6. The SAR ADC uses a switched-capacitor charge-distribution topology as in [23], asynchronous internal timing, and a CMOS sampling switch. The SAR ADC is run at a sampling rate of 400kS/s and operated at a reduced supply voltage of 1.0V to increase the T&H distortion. Note that the reduced-supply and sampling rate do not change the T&H error mechanisms, but only make them apparent to simplify the measurement setup. Since the non-linearities of the T&H switch are dependent on temperature, amplitude and frequency, the measurements were performed while varying all these parameters to ensure that the final NN captures all these dependencies and accurately captures the underlying error mechanisms. Also, it should be noted that even though the T&H distortion is of significant impact to the performance degradation, other error mechanisms are simultaneously present in the ADC core which are also targeted by the NN-based post-correction algorithm.

To characterize the ADCs for different temperature conditions, a thermal chamber was used. The temperature value fed into the network is coarsely quantized, and thus it can deviate several degrees from the actual temperature. Since the relation between the non-linearity and the temperature variation is expected to be only weakly non-linear, its sensitivity to a small temperature deviation is low. This, in turn, relaxes the design specifications for an on-chip temperature sensor required for the compensation scheme implementation in a real application scenario.

The NN was trained using measurement data from 9 frequency points, 5 input amplitudes and at 4 different temperatures. Although the non-linearity only becomes visible for

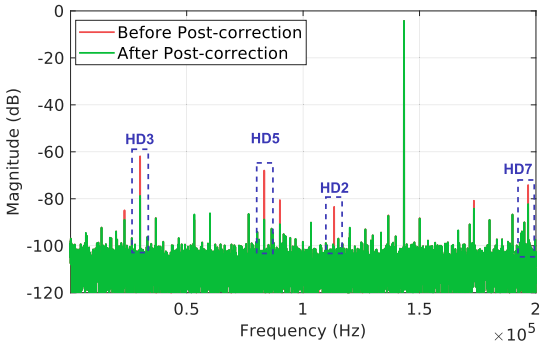


Fig. 7. FFT of signal pre- and post-correction when A_{in} = 0.7V, f_{in} = 143kHz, T = −20°C.

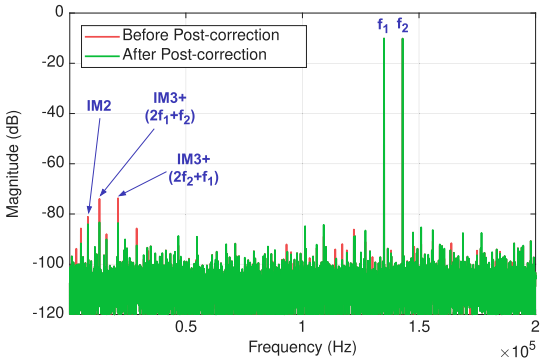


Fig. 8. FFT of signal pre- and post-correction when A_{in} = 0.35V, f_1 = 135kHz, f_2 = 143kHz, T = −20°C.

higher input amplitudes and frequencies, it is equally important to verify that the behavior of the NN is not deteriorating the input signals that do not need correction, motivating the inclusion of lower frequency and amplitude measurement points. The first subsection discusses the results after training and pruning the NN as described in Section III-B, where per-sample training was applied, indicating the maximum achievable performance using the presented NN. The second subsection discusses single sample training, where the NN is trained on a single sample, but used to correct the non-linearity for all other samples as well, to assess the generalizability of the NN.

A. Per-Sample Post-Correction

With per-sample training, the corrective ability of the neural network is tested without loss of accuracy due to the sample mismatch. An FFT of a single measurement point of one sample is shown in Fig. 7. The figure shows that the harmonic distortion is greatly reduced without introducing new spurs. The strongest harmonics are the odd-order ones, as expected from a differential ADC architecture. In Fig. 8, a two-tone test is depicted. Since the tracking error is in continuous time and the tracking error increases for larger frequencies, the strongest intermodulation components occur at the highest frequencies, i.e. the IM3 sum components (IM3+), which alias back to the band-of-interest when converted to discrete-time samples. Although the training data of the NN does not contain any two-

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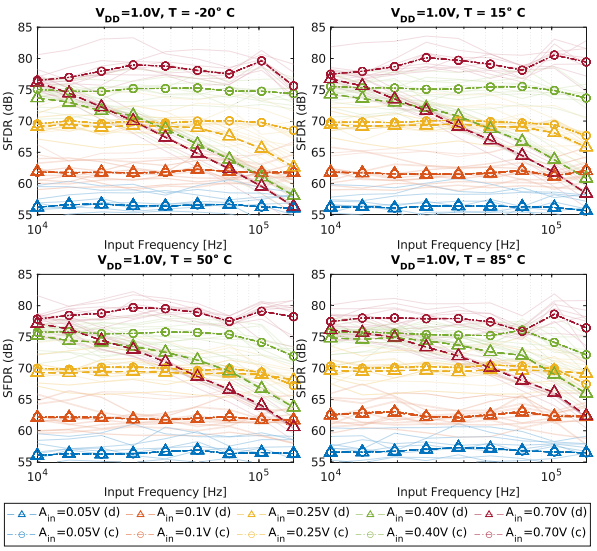


Fig. 9. SFDR pre- and post-correction of 10 silicon samples, when per-sample training is applied to the NN. In the legend, (d) denotes the distorted signal and (c) denotes the post-corrected signal.

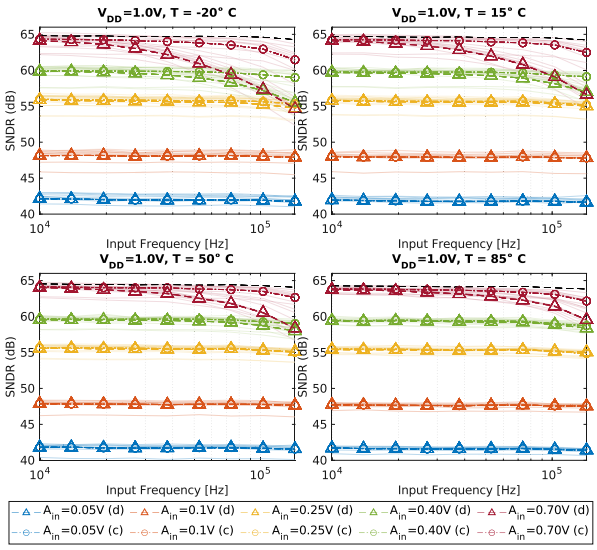


Fig. 11. SNDR pre- and post-correction of 10 silicon samples, when single-sample training is applied to the NN.

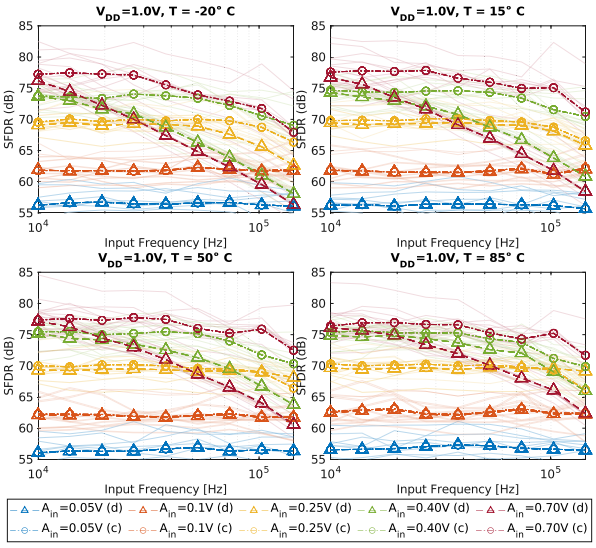


Fig. 10. SFDR pre- and post-correction of 10 silicon samples, using single-sample training.

tone input example, the NN manages to improve the SFDR by at least 8dB.

In Fig. 9, the measured SFDR of all samples are shown before (triangle markers) and after correction (circle markers): the thin lines denote the SFDR of the individual samples per input signal amplitude and the thick lines (and corresponding markers) denote the average performance of all the samples for a given input signal amplitude. It can be seen that the degradation of the SFDR is more severe at higher frequencies, higher amplitudes and at lower temperatures. This suggests that the dominant distortion mechanism is the non-linear on-resistance of the T&H switch. The proposed NN post-correction is able to restore the ADC linearity, demonstrating an SFDR improvement up to 21dB while successfully incorporating the temperature dependency which can be observed from the stable SFDR improvement over the different temperature conditions provided.

B. Per-Batch Post-Correction

To assess the generalizability of the proposed post-correction method, the NN is trained on the measurement data of a single ADC die and used for post-correction on all ADC dies. Although this will result in a slight degradation of the post-correction performance due to the mismatch between the different samples, it is more practical and cost-effective compared to the per-sample training approach. Before correction, the SFDR is limited by the third-order harmonic distortion (HD_3) for all samples; the magnitude of HD_3 varies from sample-to-sample due to device variability which means that a degradation of corrective performance is unavoidable when the NN weights are not adapted to that specific sample. The results of the ADC post-corrections using an NN trained on a single sample are shown in Fig. 10 and Fig. 11, for the achieved SFDR and SNDR, respectively. In these conditions, the NN still manages to increase the linearity of all samples. On the datapoint (i.e., frequency, amplitude, temperature combination) with the largest performance improvement, the average SFDR and SNDR improvement over all the samples is 13.3dB and 6.8dB respectively. The SFDR improvement ranges between 7dB and 20dB for the worst-case and best-case sample, respectively.

Overall, the NN-based calibration shows consistent linearity improvements when trained on a single sample, on par with previous works that calibrate each sample separately. When per-sample training is applied, the SFDR improvement surpasses the prior art, even while accounting for a much larger variety of input conditions, by varying the input signal amplitude, frequency and temperature operating point.

It should be noted that before correction, at lower input frequencies, the SNDR is not limited by T&H distortion, but instead by noise. This is highlighted in Fig. 11 by the black

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TABLE I
COMPARISON OF NON-LINEARITY CORRECTION SCHEMES IN A T&H SWITCH

	[9]	[4]	[6]	This work
Data Source	Silicon (1 sample)	Silicon (1 sample)	Silicon (1 sample)	**Silicon (10 samples)**
Method	Analytical expr.	Analytical expr.	Analytical expr.	NN
Switch Type (Enhancements)	NMOS (bootstrap, bottom-plate sampling)	NMOS (bootstrap)	NMOS	CMOS
Targeted error	Body effect	Body effect	Aperture error	All non-linearities
Training Method	Sine-wave fitting	Sine-wave fitting	Bandpass filter	Bandpass filter
Model parameters constant over	Input frequency	None	Input frequency	**Input amplitude Input frequency Temperature Samples**
Max SFDR improvement per-sample training	13dB	24dB[1]	10dB	21 dB
Max SFDR improvement single sample training	-	-	-	**13.3dB**[2]

[1] When using different coefficients for each input frequency.
[2] Average over all ADC samples.

dashed line, which shows the SNR at 0.7V input amplitude. It was also verified that the neural network does not affect the signal-to-noise ratio (SNR) of the output, ensuring that only distortion components are corrected and that the algorithm does not unintentionally modify the noise floor.

In Table I, the NN-based proposed solution is compared to prior publications on the calibration of T&H non-linearities to ensure a meaningful comparison. The prior art that reported in Table I is limited to works that include algorithm verification on silicon measurements. From the table, it can be seen that even though the ADC sampling switch used in this paper does not have analog enhancements and is therefore most strongly affected by non-linear behavior, the algorithm is able to compensate for the non-linearity over different input amplitudes, frequencies, temperatures and even different samples improving the SFDR up to 13.3dB. Note that a further increase in performance can be achieved by per-sample training, leading to a maximum SFDR improvement of 21.0dB.

V. NN INTERPRETATION AND VALIDATION

To further strengthen the confidence in the network's ability to accurately model the circuit non-idealities, the learned parameters of the network are inspected to obtain insight into the NN's internal behavior. In Section V-A, the network's weights are interpreted and in Section V-B, the learned behavior is compared against CAD simulation results.

A. NN Interpretation

As discussed in Section III-B, the learned weights tended to show many symmetries, particularly between neurons s_3 and s_4 (see Fig. 3(b)). To further uncover the meaning of this symmetry, the exact values of the weights are plotted in Fig. 12. Since the temperature and the bias value are the only components in the NN that are not input signal dependent, the combination of the two can be interpreted as a temperature-dependent bias point of the neuron, described by $c = 0.69+0.11T$ and $-c$, respectively. As illustrated in Fig. 12, the temperature-dependent bias point is pushed towards the

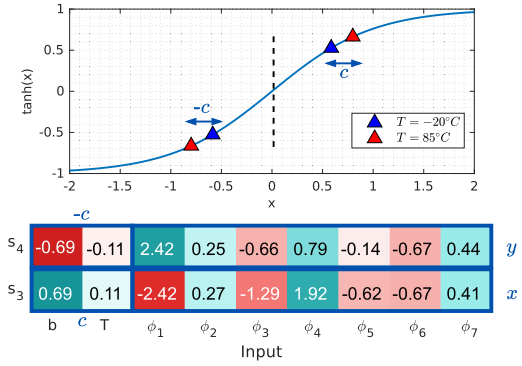


Fig. 12. The symmetric bias points of two neurons (s_3, s_4) change with temperature, at higher temperatures, the bias points move away from the linear region towards the saturation region.

saturation region of the hyperbolic tangent function at higher temperatures, whereas it remains closer to the linear region for lower temperatures.

Using the exponential definition of the hyperbolic tangent as given in (7), the sum of two non-linear neuron outputs with opposite bias can be written as:

$$\begin{aligned}\tanh(x+c)+\tanh(y-c) &= \frac{2}{e^{2(c-y)}+1}-\frac{2}{e^{2(c+x)}+1}\\ &= \frac{2\sinh(\Delta)}{\cosh(2x+2c-\Delta)+\cosh(\Delta)}\end{aligned} \tag{8}$$

where x and y denote the signal-dependent contributions to s_3 and s_4, respectively (see Fig. 12). Furthermore, the sum of the two signal-dependent terms is defined here as $\Delta = x+y$, where $\Delta \ll 1$, allowing us to approximate the $\sinh(\Delta)$ and $\cosh(\Delta)$ terms from (8) with a first-order Taylor approximation. Additionally, since $2x+2c \gg \Delta$, (8) can be further simplified to:

$$\tanh(x+c)+\tanh(y-c) \approx \frac{2\Delta}{\cosh(2x+2c)+1}. \tag{9}$$

From this equation, it is evident that the numerator term Δ is not directly dependent on the instantaneous input voltage (ϕ_1), as the ϕ_1 contribution of neurons s_3 and s_4 are canceled

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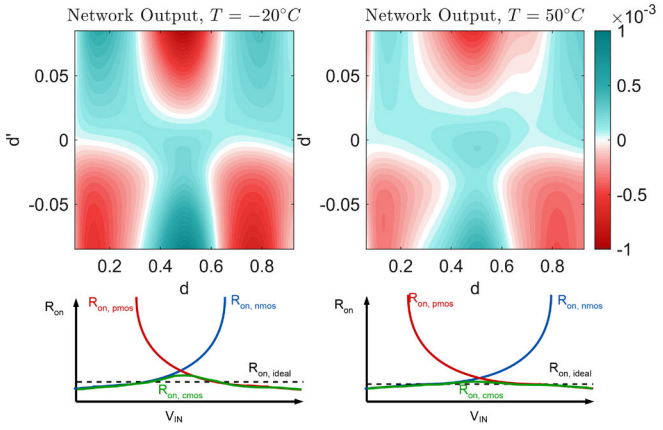


Fig. 13. Contour plot of the NN output that was trained on silicon measurement data and the relation to the on-resistance modulation for two different temperature points.

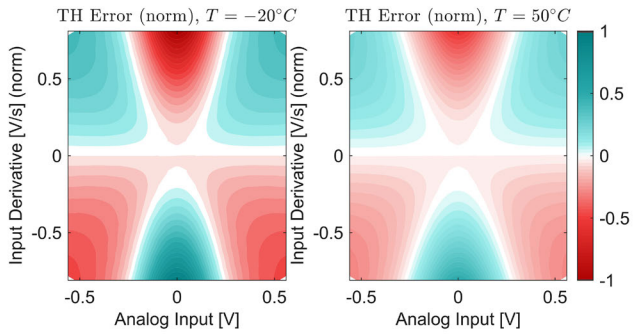


Fig. 14. Contour plot of the isolated error contribution of the non-linear T&H circuit, based on CAD simulation data for f = 40.5kHz for two different temperature points.

by the symmetric IWs; furthermore, it is also not dependent on the temperature input. In contrast, the denominator in (9) acts as a non-linear gain control and is strongly dependent on the instantaneous input signal, input frequency and temperature.

To further interpret the network behavior, the neural network outputs are plotted against the signal and derivative in Fig. 13; d represents the digital output code of the ADC normalized to the full-scale input range. Notably, the network output of the network reveals distinct symmetries. The symmetries of the network can be explained by linking the output of the network back to the dominant distortion mechanism, the non-linear on-resistance.

An artist's impression of the non-linear on-resistance for a single-ended T&H circuit is shown below the contour plot in Fig. 13. For the single-ended case, the on-resistance of each FET and the combined on-resistance is plotted against the input signal voltage. Note that the ADC measured in this paper samples differentially, meaning that the combined on-resistance modulation of the two differential input branches will be visible in the ADC output distortion. Ideally, this on-resistance is constant, depicted in Fig. 13 as $R_{on,ideal}$. To achieve this, the NN has to learn a function that applies the inverse of the non-ideal $R_{on,cmos}$; in other words, for input voltages near the edge of the input range, the magnitude of correction should be relatively large, as well as in the center of

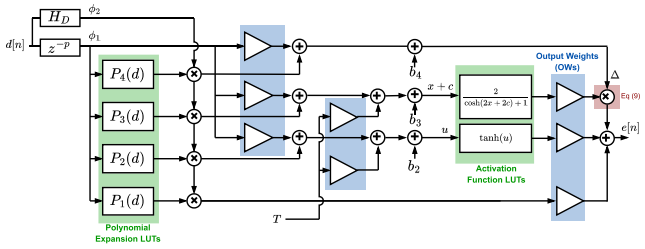


Fig. 15. Implementation diagram of the fused feature augmentation and neural network, z^{-p} is a delay line, matching the delay of the derivative filter H_D; the green blocks denote LUTs, the blue blocks multiplication with input weights (IWs) or output weights (OWs).

the input range although with opposite sign. In-between, where the $R_{on,cmos}$ crosses the average constant $R_{on,ideal}$, the correction should be at its minimum, indicated by the white areas in the contour plot. Furthermore, comparing the two contour plots of Fig 13 reveals that the magnitude of the correction decreases with increasing temperature, reflecting the reduced non-linearity of the on-resistance at elevated temperatures. This trend is consistent with the behavior illustrated in Fig. 12, confirming that the applied correction scales appropriately with temperature.

B. Cross-Validation of NN Behaviour Using CAD-Based T&H Model

To further validate the NNs output values, the T&H circuit is separately simulated (thus isolating the T&H error from the rest of the ADC) for a single frequency and 9 different amplitudes. The datapoints obtained from this simulation were interpolated to generate the contour plot shown in Fig. 14. The derivative axis was normalized to the maximum derivative of the used input signal frequency and the picture is cropped since the sinusoidal signal will yield a oval-shaped set of datapoints on the xy-plane. It can be seen that the structure of the error aligns with the results from the NN that was trained on silicon measurement data, confirming the NN's ability to accurately capture the error mechanism. Note that in the measurement data, other non-idealities of the ADC are simultaneously present and that the NN output is plotted against an approximate linear output (d), whereas the T&H error in the CAD simulation is plotted against the ideal signal thus leading to differences between the two contour plots.

To summarize, by utilizing domain knowledge in the form of input feature augmentation, it is possible to reduce the non-linearity that the neural network needs to learn and thus minimize the required number of parameters, leading to a NN containing just 5 neurons. Furthermore, the optimized network exhibits behavior that aligns closely with the dominant error mechanisms in the ADC, as confirmed through cross-validation with CAD simulations. Notably, the temperature-dependent behavior learned by the network also mirrors trends observed in simulation, reinforcing the physical relevance of the model.

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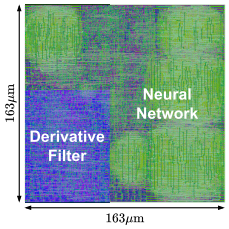


Fig. 16. Layout of the calibrator.

TABLE II
IMPLEMENTATION RESULTS

	NN	Filter
Tech. node	65nm	
Sup. voltage [V]	1.0	
Frequency [kHz]	400	
Power [uW]	3.10	2.53
Area [mm2]	0.0197	0.0068
E/cycle [pJ]	7.35	6.33

VI. IMPLEMENTATION

To evaluate the hardware cost of the proposed solution, the neural network and feature augmentation as shown in Fig. 3(a) have been combined and the network parameters have been quantized to 10-bit values. A schematic overview of the digital implementation is shown in Fig. 15. To reduce the number of multiplications, the feature augmentation and the input weights of the network have been fused into lookup tables (polynomial expansion LUTs). The ϕ_1 and T contributions are added separately, since these input features were not multiplied by ϕ_2 in the feature augmentation. To reduce the required dynamic range of the internal datapath, we directly apply the approximation from (9). This is because the two symmetrical neurons, s_3 and s_4, contribute values that are individually large but nearly cancel each other out. Their difference is small compared to their magnitudes (i.e., $x, y \gg \Delta$), so representing both accurately would otherwise require a wide datapath. A derivative filter H_D of 17 taps is used, which has anti-symmetric coefficients, halving the number of multiplications required.

This implementation is synthesized in a 65nm technology and a layout is created using an automatic place-and-route tool. The area numbers are obtained from the place-and-route tool and the power numbers are obtained using toggle-count activity obtained from post-layout simulation. The results are displayed in Table II and Fig. 16. While the hardware cost is still relatively high compared to the ADC, this is due to the high efficiency of the ADC at low supply voltage, low sampling rate and modest resolution. It is expected that for a realistic high-speed, high-resolution ADC, where the T&H error is actually a performance bottleneck, the proposed solution will become favorable in terms of cost and overall ADC performance.

VII. CONCLUSION

In this paper, a calibration solution to compensate for non-linearity in a T&H switch is presented. The calibration solution is created by embedding domain knowledge into a neural network, combining the strengths of data-driven modeling with prior knowledge. The weights of the NN were inspected while training, revealing symmetries that enabled the reduction of the network size down to just 5 neurons and allowing a high-level interpretation of the NN dynamics. The learned behavior of the NN was linked back to the dominant physical problem thereby reinforcing confidence in its corrective abilities.

An improvement up to 6.8dB and 13.3dB in the SNDR and SFDR respectively has been demonstrated over 10 different measured 12-bit SAR ADC samples using a single set of NN coefficients. When trained per sample, the SFDR improvement can be further improved up to 21.0dB. To the best of the author's knowledge, this is the first time that the T&H non-linearity has been compensated over input amplitude, frequency, operating temperature and proven with silicon measurements of multiple ADC samples and the first time that an NN-based ADC calibration solution is interpreted and linked back to the physical error mechanism.

VIII. FUTURE WORK

The modular nature of the NN architecture allows for extension by incorporating additional sensory inputs. For instance, variations in supply and reference voltage of the ADC could be integrated into the model to further enhance calibration accuracy or broaden the range of supported operating conditions. Moreover, future research could explore whether the difference between the samples can be correlated to a measurable quantity such as process variation indicators, further increasing the performance while avoiding the need for per-sample training. While this work focused on the T&H circuit due to the many factors affecting this distortion mechanism, the methodology can be extended and applied to other ADC architectures suffering from different non-linear behavior.

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